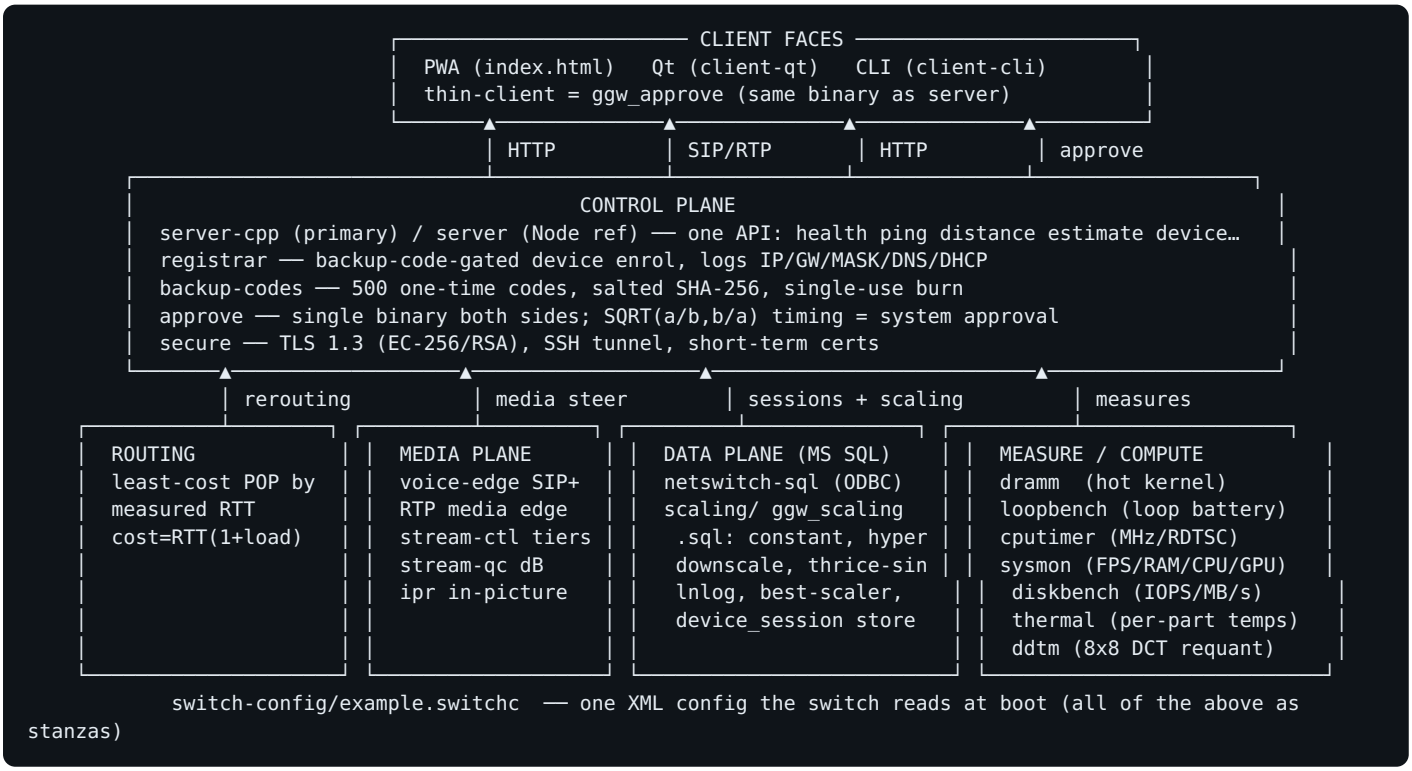
# ARCHITECTURE

## 6GGW / NetSwitch — architecture & component map

One product: a serverless installable app plus a suite of native C++ modules and an MS SQL data plane. Everything below speaks one HTTP/SIP/SQL wire, so the four client faces (PWA, Qt, CLI, thin-client) are interchangeable. This document is the box graph, the data flows, and a catalogue of every component and its files. Companion docs: `CLIENTS.md` (prose per-component), `MATH-INDEX.md` (every math function, numbered).

### Box graph




### The end-to-end flow (a device joining and making a call)

```
1. user opens the thin client (PWA / ggw_approve)
2. REGISTER with a one-time backup code   — backup-codes verifies + burns it
3. registrar logs the device               — IP/GW/MASK/DNS/DNS2/DNS-periph/DHCP  → MS SQL device_session
4. approve                                  — same binary on server+client, timing ratio ~1.0 = APPROVED
5. rerouting engine picks the edge          — measured RTT, cost = RTT·(1+load/100)
6. SIP INVITE → voice-edge media edge       — RTP (G.711 μ-law) relayed at the chosen POP
7. SCCM block scaled                        — ddtm 8x8 DCT + requant; MS SQL scaling picks best scaler
8. stream-ctl / stream-qc                   — bitrate tiers, pipe floor, dB pixelation gate
```

### Component catalogue

| # | Component | Kind | Primary file(s) | What it is |
|---|---|---|---|---|
| 1 | `index.html` `sw.js` `manifest` | PWA | web | installable phone app base |
| 2 | `server-cpp` | C++ | `ggw_server.cpp` | primary native server; the one API |

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| | | | | |
|---|---|---|---|---|
| 3 | `server` | Node | `server.js` `backup.js` `restore.js` | reference server, same API |
| 4 | `client-qt` | Qt C++ | `main.cpp` | desktop/mobile client, 4 tabs |
| 5 | `client-cli` | C++ | `ggw_cli.cpp` | CLI client (Win/Lin/mac) |
| 6 | `secure` | C++/OpenSSL | `ggw_secure.cpp` | TLS 1.3, SSH tunnel, short-term certs |
| 7 | `dramm` | C++/CUDA | `cpu_benchmark.*` `ggw_dramm.cpp` | tight compute kernel + benchmark |
| 8 | `diskbench` | C++ | `ggw_diskbench.cpp` | seq/random disk IOPS + MB/s + latency |
| 9 | `cputimer` | C++ hdr | `cpu_mhz_timer.h` `ggw_cputimer.cpp` `ggw_radiosig.cpp` | RDTSC MHz timer; radio-param reduction |
| 10 | `stream-qc` | C++ | `ggw_streamqc.cpp` | PSNR/blockiness pixelation watch (dB) |
| 11 | `stream-ctl` | C++ | `ggw_streamctl.cpp` | bitrate tiers, pipe-fill, battery, split |
| 12 | `ipr` | C++ | `ggw_ipr.cpp` | image-in-picture compositor |
| 13 | `netswitch-sql` | C++/ODBC | `netswitch_sql.cpp` | MS SQL client, forced pw change, timed RTT |
| 14 | `netswitch-sql/scaling` | T-SQL | `ggw_scaling.sql` | scaling math native in MS SQL (see below) |
| 15 | `ui-panels` | HTML | `index.html` | 4 operator UI panels |
| 16 | `thermal` | C++ | `ggw_thermal.cpp` | per-part temps, throttle watch |
| 17 | `loopbench` | C++ | `ggw_loopbench.cpp` | loop battery; optimized-arith + push |
| 18 | `sysmon` | C++ | `ggw_sysmon.cpp` | live FPS/ROM/RAM/CPU%/GPU% monitor |
| 19 | `backup-codes` | C++ | `ggw_backupcodes.cpp` | 500 one-time codes, salted SHA-256 |
| 20 | `registrar` | C++ | `ggw_registrar.cpp` | code-gated device enrol + net log |
| 21 | `voice-edge` | C++ | `ggw_voice_edge.cpp` | SIP register + RTP through a media edge |
| 22 | `ddtm` | C++ | `ggw_ddtm.cpp` | 8x8 DCT, powered/unpowered, requant scaler |
| 23 | `approve` | C++ | `ggw_approve.cpp` | single-binary system-approval by timing |
| 24 | `switch-config` | XML | `example.switchc` | the one boot config, all modules as stanzas |

## Portability (server choices)

- **Data plane** targets **Microsoft SQL Server 2026** (T-SQL, tested syntax); the scaling functions use only standard

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SQL maths, so an Oracle/PostgreSQL port is a small dialect change ( `LOG` / `POWER` / `SIN` exist in all three).
- **HTTP front** is server-agnostic: `server-cpp` runs standalone, or sit it behind **Nginx / Apache** as a reverse proxy — no code change.
- Every C++ module builds on Linux (g++) and Windows (mingw/MSVC); the phone builds come from the same sources with the NDK / Qt mobile kits.

## Where the money-math lives

`120 W` is the fixed power reference in the DDTM/scaling constant ( `ggw_scaling.sql` → `ddtm_constant_calc` , `/ @watts` default `120.0` ). It is present and set to 120 W as specified. ```

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